Markovian Approach to Model User Behaviour in E-Commerce

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Outline

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Introduction and Problem Description

Main Goal: Build a probabilistic Markov Chain model of user journeys on e-commerce websites

Key Objectives:

- Analyze clickstream data to map common user paths
- Distinguish sequences leading to purchase vs. abandonment
- Provide actionable insights for e-commerce businesses

Business Value

Understanding user journeys enables businesses to:

1. Strengthen profitable user paths and fix navigational dead ends
2. Predict next likely clicks and personalize experiences in real-time
3. Pinpoint pages where users are most likely to abandon
4. Suggest products from categories common in successful purchase journeys

Why Markov Chains?

Challenges with Raw Clickstream Data:

- Extremely noisy
- Average metrics fail to capture sequential decision-making
- Users deliberate over multiple categories before purchasing

Our Approach:

- Model conversion intents (not just sentiment)
- Focus on journeys ending in purchases or bounces
- Reconstruct linearity via temporal sorting within sessions
- Build probabilistic framework: $P(\text{next state} \mid \text{current state})$

Data Source

Dataset: E-Commerce Behaviour Data from Multi-Category Store (Kaggle)

Key Features Used:

- `user_session`: Group events by browsing session
- `event_time`: Sort events chronologically
- `event_type`: Identify actions (view, cart, purchase)
- `category_code`: Identify states/pages (e.g., electronics.smartphone)

Scope:

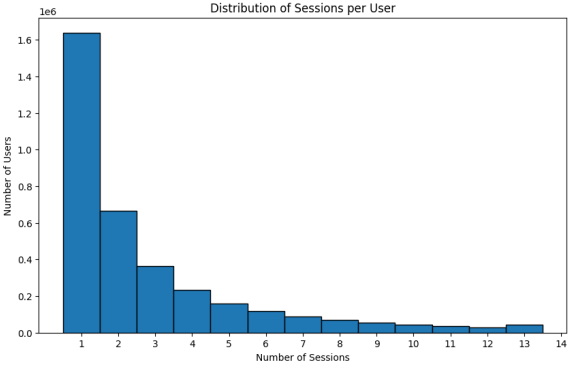
- November 2019 data: 67 million click events
- $\sim$4 million unique users
- $\sim$14 million unique sessions
- Limited to 100,000 sessions for analysis (computational constraints)

User Session Distribution

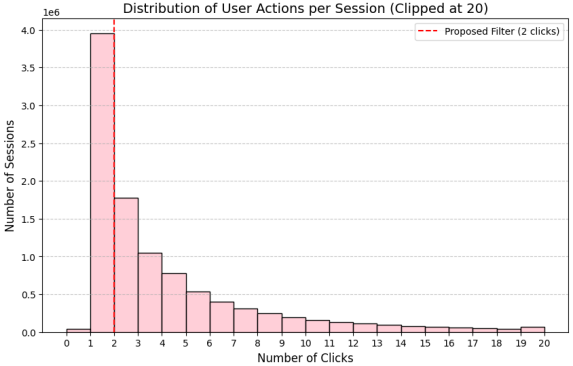
Sessions per User:

- Mean: 3.73 sessions
- Median: 2 sessions
- Distribution: Highly right-skewed
- Return rate: $\sim$50%

Note: Most users browse 1-2 times; small portion are highly engaged repeat visitors



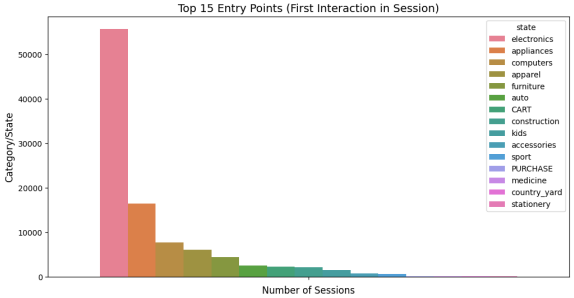
Clicks per Session Distribution



Key Observations:

- Peak at 2 clicks (high noise)
- Many sessions are brief interactions
- Users bounce after 1-2 clicks
- Justifies filtering sessions < 3 interactions

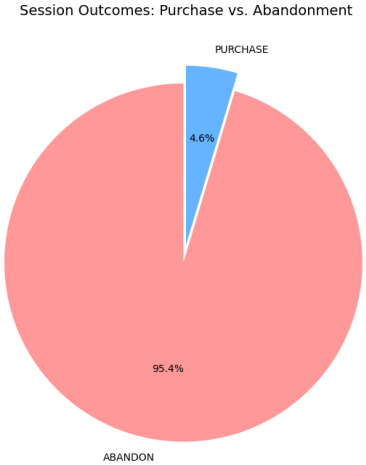
Top 15 Entry Points



Insights:

- Electronics dominates with highest entry traffic
- Computers and appliances follow
- Stationery, country_yard and medicine have the lowest entry rates

Purchase vs. Abandonment Patterns



Key Findings:

- Majority: No purchase (abandonment)
- Window shoppers/recreational browsers are probably dominating
- Some add to cart but don't complete (may come for another session)
- Multiple purchases possible per session (one order)

Mathematical Context

State Space:

$$S = \{s_{\text{start}}, s_{\text{cart}}, s_{\text{purchase}}, s_{\text{abandon}}, c_1, c_2, \ldots, c_n\}$$

where c_i represent aggregated product categories

Markov Property:

- First-Order: $P(X_{n+1} \mid X_n, X_{n-1}, \ldots) = P(X_{n+1} \mid X_n)$
- Second-Order: $P(X_{n+1} \mid X_n, X_{n-1}, \ldots) = P(X_{n+1} \mid X_n, X_{n-1})$
- Model as Discrete-Time Markov Chain (DTMC) based on clicks

Data Processing Pipeline

Starting Point: 67 million clicks from November 2019

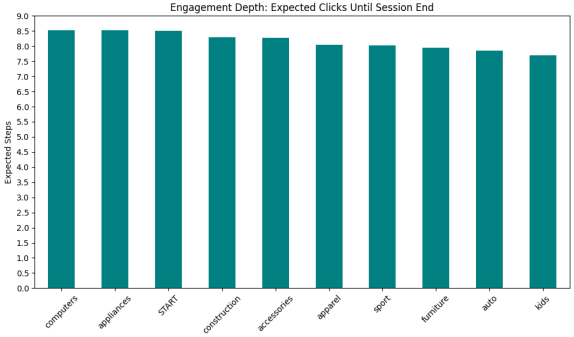
Steps:

1. **Preprocessing:**
   - Convert time to datetime format
   - Drop clicks with missing category codes
   - Aggregate subcategories (e.g., electronics.smartphone $\rightarrow$ electronics)
   - Organize by session ID ($\sim$14 million unique sessions)
2. **State Space Engineering:**
   - Override categories with high-value actions (CART, PURCHASE)
   - Append START at beginning and ABANDON at end (if no purchase)
   - Result: Session ID + ordered list of states/transitions

Methodology (continued)

3. **Noise Filtering:**
   - Remove sessions with < 3 interactions
4. **First-Order Modeling:**
   - Construct 17×17 Transition Probability Matrix (TPM)
   - States: START, ABANDON, CART, PURCHASE + 13 product categories
   - Visualize using heatmap
   - Calculate sparsity: $\sim 12.5\%$
5. **Absorption Analysis:**
   - Identify transient vs. absorbing states
   - Calculate "Expected Steps to Absorption" via fundamental matrix

Expected Steps to Absorption



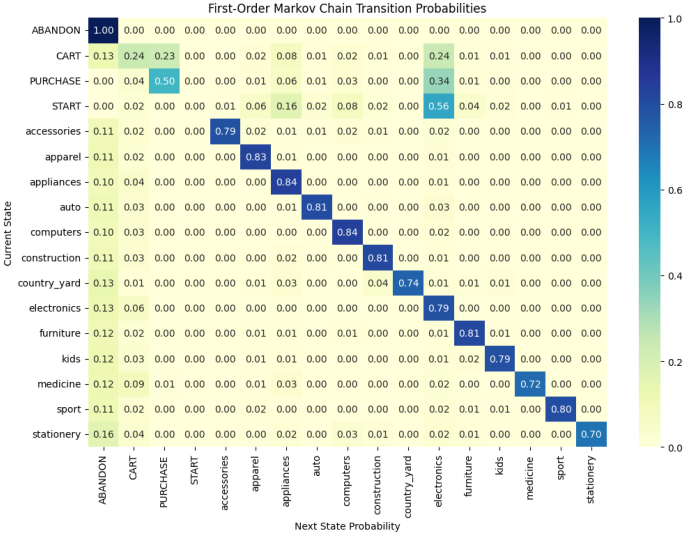
Interpretation:

- Measures engagement depth per category
- Higher values = more browsing before purchase/abandon
- Fundamental matrix calculation shows category stickiness

Advanced Modeling

6. **Clustering:**
   - Segment customers into types based on behavior features
   - Goal: separate models per customer type $\rightarrow$ better predictive power
   - Explored only: compute limits blocked per-cluster models
7. **Continuous-Time Markov Chain (CTMC), proposed extension:**
   - DTMC limitation: Treats all transitions equally regardless of duration
   - CTMC: Differentiates urgency vs. deliberation
   - Uses sojourn times (how long user stays in state i)
   - Rate matrix: λ_{ij} = (number of transitions i $\rightarrow$ j) / (total time spent in i)

Transition Probability Matrix (TPM) Heatmap



Key Observations from TPM

From the Transition Probability Matrix heatmap:

- If user clicks in a category, next click likely in **same category**
- From START state, most probable next state is **electronics**
- ABANDON is an **absorbing state**
- From CART: can stay in cart, purchase, abandon, or browse category
  - If browsing from cart, most likely **electronics**
- From PURCHASE: 50% probability stays in purchase, next likely **electronics**
  - Multiple purchases possible within single session (one order)

First-Order vs. Second-Order Model Comparison

Metric	First-Order	Second-Order
Sparsity	12.5%	22.8%
Parameters (k)	Lower	10× Higher
Log-Likelihood (LL)	−60 million	−111 million
AIC	Lower (Better)	Higher (Worse)
BIC	Lower (Better)	Higher (Worse)

Winner: First-Order Model

Model Selection Metrics Explained

Sparsity (12.5% vs. 22.8%):

- Percentage of TPM containing zeros
- Higher order $\rightarrow$ exponentially more combinations $\rightarrow$ more sparsity

Parameters (k):

- Model complexity (degrees of freedom)
- Second-order $10\times$ more complex without accuracy improvement

Log-Likelihood (LL):

- Measures how likely observed data is given the model
- Closer to 0 = better fit
- First-order: -60M (better) vs. Second-order: -111M (worse)
- Second-order sparse matrix penalizes unseen sequences heavily

Information Criteria

Akaike Information Criterion (AIC):

$$\text{AIC} = 2k - 2\ln(\hat{L})$$

- Balances accuracy vs. complexity
- Lower is better
- First-order has much lower AIC

Bayesian Information Criterion (BIC):

$$\text{BIC} = k\ln(n) - 2\ln(\hat{L})$$

- Harsher penalty for parameters (especially with large n)
- Lower is better
- First-order strongly preferred
- Second-order complexity mathematically unjustifiable

Model Testing and Robustness

Temporal Consistency Check:

- Compared Week 1 vs. Week 4 of November
- Similar click volumes: 7.8M vs. 8M

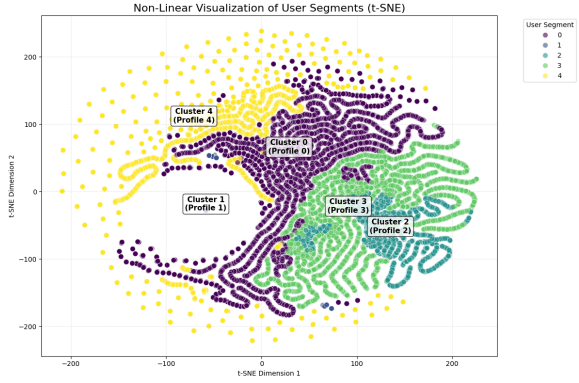
Behavioral Drift Analysis:

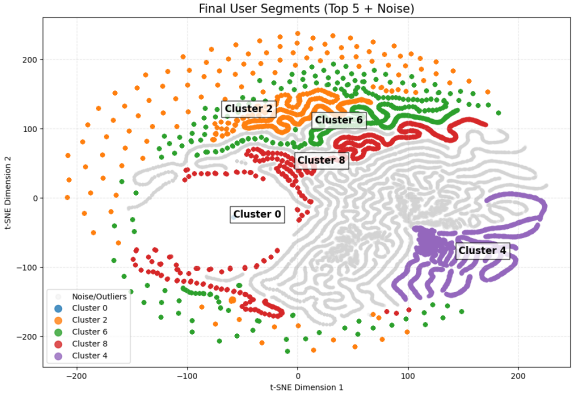
- Built separate TPMs for Week 1 and Week 4
- Calculated Frobenius norm to measure drift
- Result: Drift $= 1.22$ (threshold: 3)

Implications:

- User transition probabilities remained stable
- Early month data reliably predicts late month behavior
- Category-based predictions are trustworthy
- Top-3 hit rate of 87%+ for every state, but CART Top-1 recall is 0%

Customer Clustering

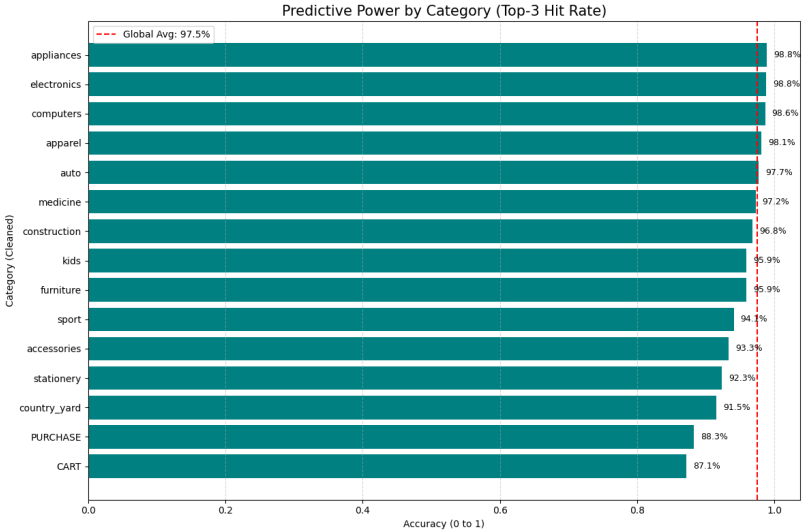




Segmentation Aims:

- Separate CTMC models for each customer type
- Hypothesis: Improved predictive power through segmentation

Model Predictive Power



Prediction Accuracy by Category

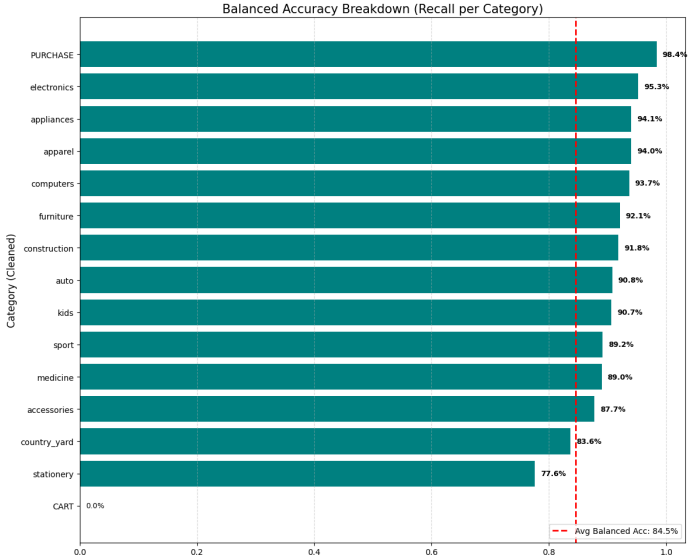
High Accuracy Categories (more than 95%):

- Product categories perform exceptionally well
- Electronics, appliances, computers, apparel all $>98\%$
- Indicates strong predictive patterns in browsing behavior

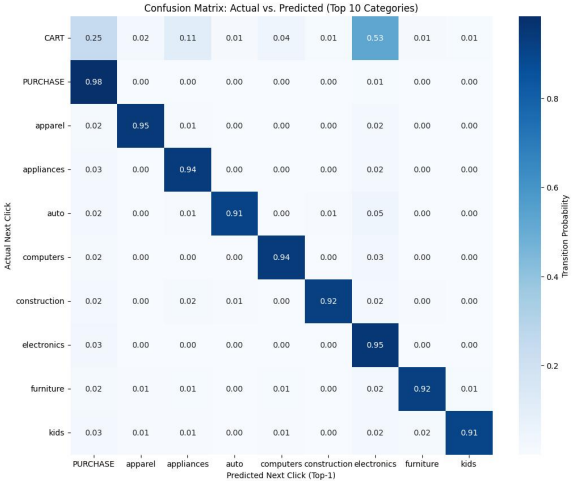
Lower Accuracy Categories:

- CART (87.1%): Complex decision point with multiple paths
- PURCHASE (88.3%): Terminal action with varied origins
- Country_yard (91.5%), Stationery (92.3%), Accessories (93.3%)
- Top-3 hides imbalance: CART is never the Top-1 prediction (0% recall)

Balanced Accuracy By Category



Confusion Matrix: Prediction Performance



Confusion Matrix Insights

Notable Patterns:

- **CART confusion:** 53% predicted as electronics (next likely browse)
- 25% predicted as PURCHASE; CART itself is never predicted
- High cross-category confusion minimal (mostly $<5\%$)

Model Strengths:

- Clear separation between states (minimal off-diagonal noise)
- High-value actions (PURCHASE) nearly perfectly predicted
- Category transitions follow logical browsing patterns

Validation: Session-split validation ensures model generalizes to unseen data

Model Limitations

1. **Memoryless Constraint (Markov Property):**
   - First-order model only considers current state
   - Cannot distinguish: 20-item browser vs. new category entrant
   - User narrative compressed into single step
   - Higher-order models too sparse (shown by AIC/BIC)

2. **Loss of Granularity (State Aggregation):**
   - Subcategories aggregated to root (electronics.smartphone $\rightarrow$ electronics)
   - Stabilizes math but loses product intent
   - Predicts "electronics" purchase, not "phone vs. TV"

Additional Limitations

3. Temporal Blindness (Initial DTMC):

- All transitions treated as equal time steps
- 10-minute dwell (high intent) = 1-second bounce (low intent)
- CTMC extension proposed as future work

4. Stationarity Assumption:

- Static TPM assumes constant behavior throughout November
- Fails to capture event-driven shifts (e.g., Black Friday)

5. Data Exclusion Bias:

- Removed "unknown" categories and sessions ≤ 2 interactions
- Survivor bias: Model optimized for engaged users
- Less effective for cold-start users or high-bounce traffic

Conclusions

Key Achievements:

- Successfully built first-order Markov Chain model for e-commerce user behavior
- Demonstrated superiority of first-order model over second-order (AIC/BIC)
- Identified key transition patterns and absorbing states
- Top-3 accuracy 97.45% (87%+ for every state); balanced accuracy 84.5%

Practical Insights:

- Electronics dominates user journeys
- Most users are window shoppers (high abandonment)
- Cart state is critical decision point and is under-represented
- Behavior stable across time periods (low drift)

Future Directions

Model Enhancements:

- Incorporate contextual features (time of day, seasonality, promotions)
- Hybrid models combining Markov with machine learning
- Dynamic TPM updates for event-driven behavior changes
- Personalized transition matrices per user segment

Applications:

- Real-time intervention systems (prevent abandonment)
- Personalized product recommendations
- A/B testing navigation structures
- Customer lifetime value prediction

References

E-Commerce Behaviour Data from Multi-Category Store
Kaggle Dataset
Available at: https://www.kaggle.com/

Thank You!

Questions?